

Jaypee Institute of Information Technology, Noida



Project Title: Link Analysis on Vote Wiki Dataset

Name of Student	Enrollment No.
Ishita Agarwal	2503030003
Parnika Jain	2503030005
Apurva Ananad	2503030007

Course Name: Web Intelligence

Program: Mtech (CSE)

2nd Sem

1. Abstract

This project focuses on performing link analysis on the Wiki-Vote dataset to identify influential nodes (users) in a network. The dataset is modeled as a directed graph where nodes represent users and edges represent voting relationships. Two fundamental link analysis algorithms-PageRank and HITS (Hyperlink-Induced Topic Search) are implemented to evaluate node importance. Additionally, structural properties such as indegree and outdegree are computed for each node. The system allows users to input a node ID and retrieve its corresponding metrics, including PageRank score, HITS authority and hub values, and degree information. This project demonstrates how graph-based ranking algorithms can be applied to social network data to uncover influence patterns and user behavior.

2. Introduction

With the rapid growth of online platforms, analyzing relationships between users has become crucial for understanding influence, recommendation systems, and ranking mechanisms. Link analysis is a technique used in graph theory and data mining to evaluate the importance of nodes based on their connections.

One of the most well-known applications of link analysis is the Google Search engine, which uses PageRank to rank web pages. Similarly, algorithms like HITS identify authoritative and hub nodes in a network.

In this project, we apply these techniques to the Wiki-Vote dataset, which represents voting interactions among Wikipedia users. By modeling the dataset as a directed graph, we analyze how influence flows through the network and how users can be ranked based on their connectivity and interaction patterns.

3. Literature study

3.1 The anatomy of a large-scale hypertextual Web search engine [1]

This paper explains how Google was designed as a powerful web search engine. The main idea of the paper is to improve search results by using the link structure of the web instead of only matching keywords. It introduces the PageRank algorithm, which ranks web pages based on how many important pages link to them, making results more relevant and high quality. The paper also describes how Google efficiently crawls, stores, and indexes millions of web pages, and uses features like anchor text and word proximity to give better search results. Overall, the main work of the paper is building a scalable and efficient search engine that gives more accurate results than earlier systems.

3.2 Link Analysis: Hubs and Authorities on the World Wide Web [2]

This explains how the HITS algorithm works for ranking web pages. The main idea is that web pages can be of two types: hubs (pages that link to many useful pages) and authorities (pages that contain valuable information). The paper studies how these two types support each other good hubs point to good authorities, and good authorities are linked by good hubs. It also gives a mathematical analysis (using matrices and probability) and shows that, in simple terms, HITS ranking is closely related to counting incoming and outgoing links. Overall, the main work of the paper is to analyze and explain how HITS ranks web pages using link structure.

3.3 Personal vs. know-how contacts: which matter more in wiki elections? [3]

This paper studies how different types of social connections affect success in Wikipedia elections. It compares personal contacts (direct connections like friends or immediate neighbors) and know-how contacts (indirect or influential connections in the network). The main finding is that personal contacts are more important for getting selected as an administrator and for active participation in voting. People who have more direct connections are more likely to win elections. However, know-how contacts also help in spreading information and influencing voting, but their impact is smaller compared to personal contacts. Overall, the paper shows that direct relationships matter the most in online election success.

4. Dataset

The Wiki-Vote dataset is obtained from the Stanford Network Analysis Project. It contains voting data from Wikipedia administrator elections.

Key Characteristics:

- Nodes represent users
- Directed edges represent votes ($A \rightarrow B$ means user A voted for user B)
- Graph type: Directed and unweighted
- Real-world social network data

Dataset Statistics:

- Total Nodes: ~7,000+
- Total Edges: ~100,000+
- Represents administrator elections and voting behavior
- Includes both successful and unsuccessful elections

This dataset is ideal for link analysis because:

- It captures real user interactions
- It reflects influence and trust relationships
- It contains complex network structures like hubs and authorities

5. Methodology

5.1 Graph Representation

The dataset is converted into a directed graph:

- Each user $\rightarrow$ node
- Each vote $\rightarrow$ directed edge
- Stored using adjacency lists for efficiency

We compute:

- Indegree: Number of incoming edges (votes received)
- Outdegree: Number of outgoing edges (votes given)

5.2 PageRank Algorithm

PageRank measures the importance of a node based on incoming links. A node is considered important if it is linked by other important nodes.

Key Idea:

- A “random surfer” model distributes importance across the graph
- Uses iterative computation until convergence

Steps:

1. Initialize rank values for all nodes
2. Iteratively update ranks based on incoming links
3. Apply damping factor (usually 0.85)
4. Stop when convergence threshold is reached

Outcome:

- Each node gets a PageRank score
- Higher score $\rightarrow$ more influential node

Formulation:

PageRank is a function that assigns a real number to each page in the Web (or at least to that portion of the Web that has been crawled and its links discovered). The intent is that the higher the PageRank of a page, the more “important” it is. In PageRank, each website is represented by a node in a directed graph, and links between websites are represented as directed edges. PageRank assume that users behave like a random surfer on any websites. At each step, random surfer has two options: following a link at random with probability β , or jumping to some random page with probability $1-\beta$.

The rank of node j can be computed by the following PageRank equation [2] (see Eq. (1)):

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{N} \quad (1)$$

In Eq. (1), constant β is called as the tax rate, which describe the probability for a user to go to another website following a link on one website. Usually β is equal to $0.8 \sim 0.9$. N is the total number of websites in database, and d_i is the outdegree of node i .

The Google Matrix is defined by the following equation (see Eq. (2)):

$$\mathbf{A} = \beta \mathbf{M} + (1 - \beta) \left[\frac{1}{N} \right]_{N \times N} \quad (2)$$

The rank vector $\mathbf{r}$ can be solved by the following iterative update equation (see Eq. (3)) with random initialization:

$$\mathbf{r}_{new} = \beta \mathbf{M} \cdot \mathbf{r}_{old} + \left[\frac{1 - \beta}{N} \right]_N \quad (3)$$

5.3 HITS Algorithm

HITS assigns two scores to each node:

- Authority Score → how valuable the node is (receives links)
- Hub Score → how well the node points to good authorities

Key Idea:

- Good hubs point to good authorities
- Good authorities are pointed to by good hubs

Steps:

1. Initialize hub and authority scores
2. Update authority scores from incoming links
3. Update hub scores from outgoing links
4. Normalize values
5. Repeat until convergence

Outcome:

- Each node has:
 - Authority score
 - Hub score

5.4 Node Query System

A user-friendly system is implemented where:

- User inputs a node ID
- System returns:
 - Indegree
 - Outdegree
 - PageRank score
 - HITS authority score

- HITS hub score

This makes the analysis interactive and helps in understanding individual node importance.

5.5 Tools & Technologies Used

- Programming Language: Python
- UI - Streamlit
- Libraries:
 - NetworkX (for graph processing)
 - NumPy (for numerical computation)
- Data Structure:
 - Adjacency List

6. Results

The implemented system successfully analyzed the Wiki-Vote dataset using PageRank, HITS, and Degree Centrality. The results highlight both strong correlations and notable differences between ranking methods.

6.1 User Interface

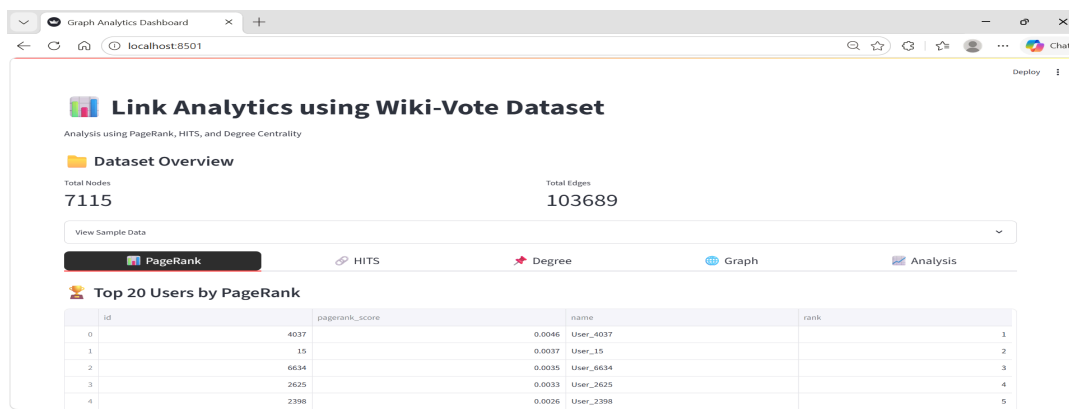


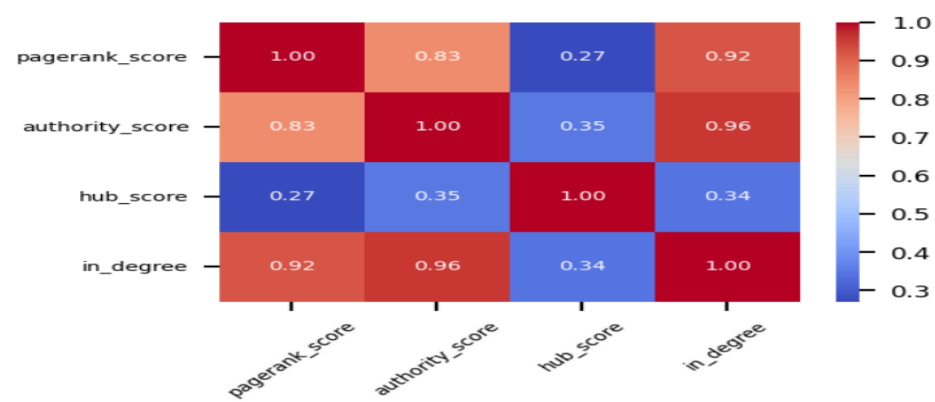
Figure 1 User Interface

6.2 Correlation Analysis

To understand the relationship between different ranking metrics, Pearson correlation coefficients were computed:

Table 1: Correlation Analysis Between Network Centrality Metrics

Metric Pair	Correlation (r)	Interpretation
PageRank vs Authority	0.835	Strong positive correlation
PageRank vs In-Degree	0.923	Very strong correlation
Authority vs In-Degree	0.960	Extremely strong correlation
Hub vs In-Degree	0.340	Weak to moderate correlation
Hub vs PageRank	0.271	Weak correlation



Key Findings:

- The highest correlation (0.960) is observed between Authority score and In-Degree, indicating that nodes receiving more votes tend to be strong authorities.
- PageRank and In-Degree (0.923) also show a strong relationship, confirming that PageRank is influenced by incoming links, but not solely dependent on them.
- PageRank vs Authority (0.835) demonstrates that both algorithms capture node importance, though through different mechanisms.
- Hub scores show weak correlation with other metrics, indicating that hubs represent a different role in the network (active linking rather than influence).

6.2 Top-K Overlap Analysis

To compare ranking consistency at the top level, the overlap between top-ranked nodes was measured.

- For $K = 10$, the overlap between PageRank and Authority rankings is:

Overlap = 40%

- This means that 4 out of the top 10 nodes are common between the two methods.

Interpretation:

- Although overall correlation is high, the relatively low Top-K overlap shows that:
 - Different algorithms prioritize different nodes at the top
 - Ranking differences become more prominent for highly influential nodes

6.3 Graph Structure Observations

- High-ranking nodes tend to form densely connected subgraphs
- The network exhibits skewed distribution, where a small number of nodes dominate influence
- Presence of hub-like nodes that actively vote but are not necessarily influential themselves

7. Conclusion

This project demonstrates the effectiveness of link analysis algorithms in identifying influential nodes within a real-world social network.

- The strong correlation between Authority and In-Degree ($r = 0.960$) confirms that authority is largely driven by the number of incoming links.
- PageRank ($r = 0.923$ with In-Degree) extends this by incorporating the importance of linking nodes, making it a more robust global ranking method.
- The relatively weak correlation of Hub scores ($r \approx 0.27\text{--}0.34$) highlights their distinct role in identifying nodes that actively connect to others rather than those that are influential.

8. References

- [1] L. Page, S. Brin, R. Motwani, and T. Winograd, “The PageRank Citation Ranking: Bringing Order to the Web,” Stanford InfoLab, Technical Report, 1999.
- [2] C. H. Q. Ding, X. He, H. Zha, P. Husbands, and H. D. Simon, “Link Analysis: Hubs and Authorities on the World Wide Web,” *SIAM Review*, vol. 46, no. 2, pp. 256–268, 2004.
- [3] S. Brin and L. Page, “The Anatomy of a Large-Scale Hypertextual Web Search Engine,” *Computer Networks and ISDN Systems*, vol. 30, no. 1–7, pp. 107–117, 1998.
- [4] Dataset: Wiki Vote , Link: <https://www.kaggle.com/datasets/tananrules/wiki-vote/code>

